

Eenheid 1.10

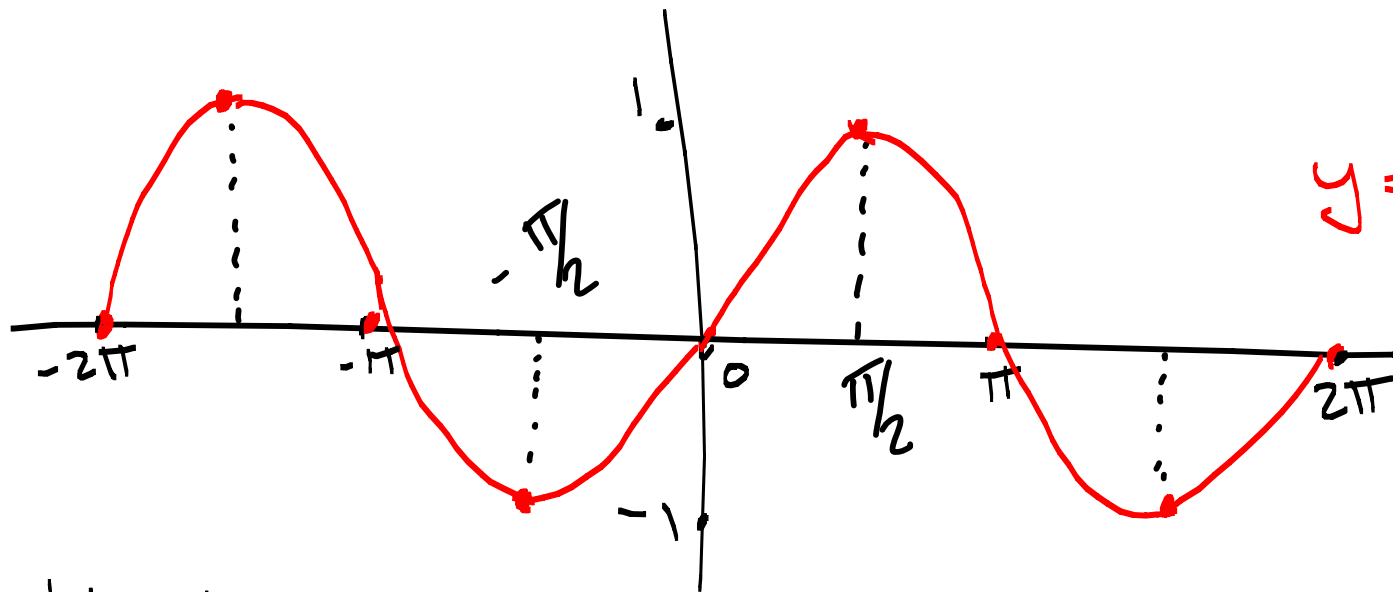
Note Title

2016/02/21

# Inverse trigonometrische functies

Stewart bl. 63-66

$$\underline{y = \sin x}$$



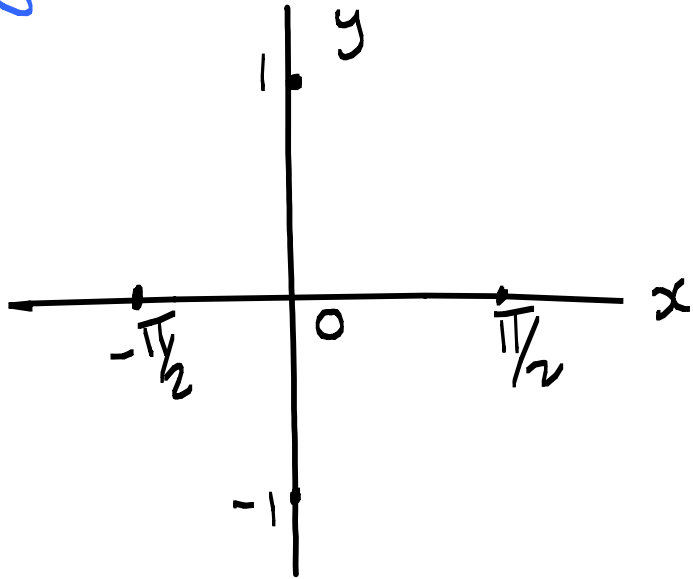
$$y = \sin x$$

Het  $y = \sin x$  is inverse functie??

Ja / Nee? Motiveren!

Plan? Inverse??

•  $y = \sin x$

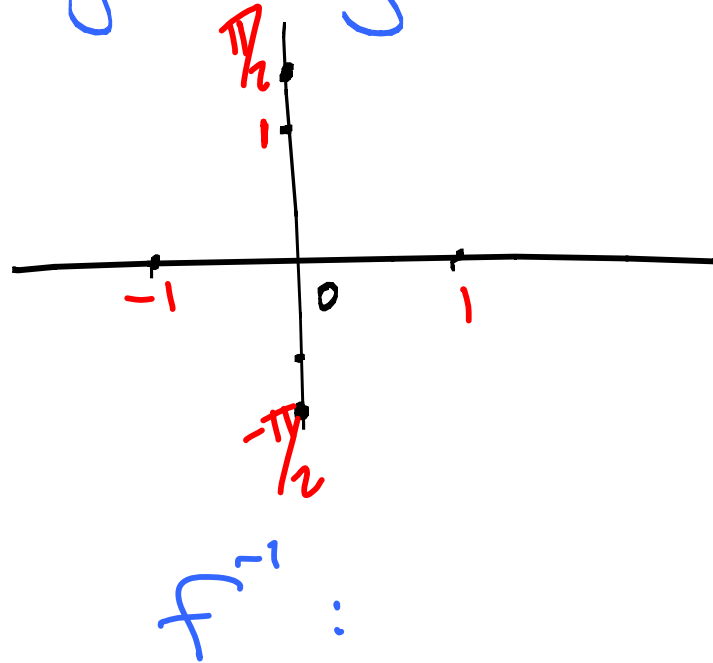


$f$ :

$$y = \text{bg} \sin x = \arcsin x = \sin^{-1} x \neq \frac{1}{\sin x}$$

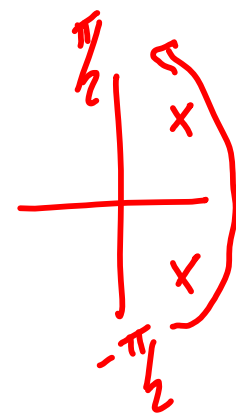
(Notatie)

•  $y = \text{bg} \sin x$



$f^{-1}$ :

Definisie:  $y = \text{bg} \sin x \Leftrightarrow \sin y = x$   
(Volledig) en  $y \in [-\pi/2, \pi/2]$



Kansellasie Vergelykings:

1.  $\sin(\text{bg} \sin x) = x$  as  $x \in$

2.  $\text{bg} \sin(\sin x) = x$  as  $x \in$

Los op:

①  $\sin x = -1/2$

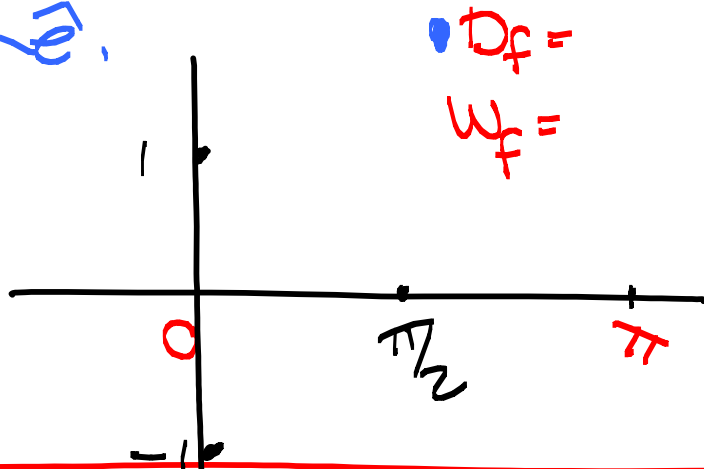
$$\textcircled{2} \quad y = \log \sin \left(-\frac{1}{2}\right)$$

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$$\textcircled{3} \quad \sin^{-1} \left( \sin \frac{7\pi}{3} \right) = y$$

- $y = \cos x$  moet een-tot-een wees om inverse funktie te hē.

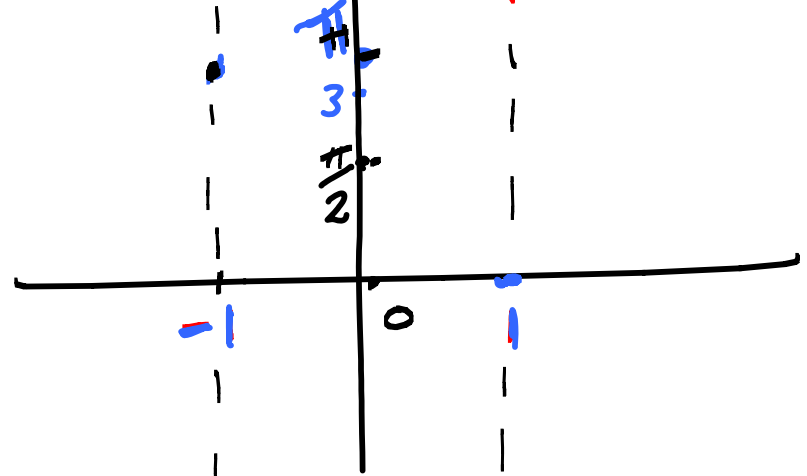
Plan:



$$y = \lg \cos x$$

$$D_{f^{-1}} =$$

$$W_{f^{-1}} =$$



\* Definitie:  $y = \cos^{-1} x \Leftrightarrow \cos y = x$  en  $y \in [0, \pi]$

Kansellatie vergelijkingen:

$$1. \cos(\lg \cos x) = x \quad \text{vir } x \in$$

$$2. \lg \cos(\cos x) = x \quad \text{vir } x \in$$

① Log op:  $y = \lg \cos(0.5)$

Vind: ②  $y = \lg \cos\left(-\frac{\sqrt{3}}{2}\right)$

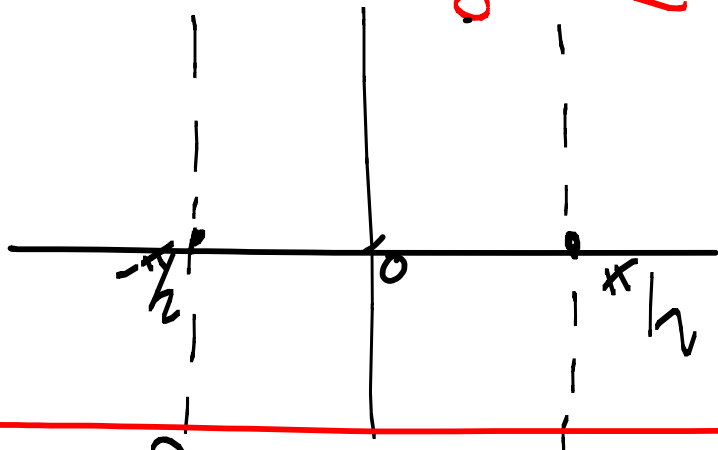
$$\textcircled{3} \quad y = \log \cos \left( \cos \left( -\frac{\pi}{2} \right) \right)$$

of Ty kan ook die kansellasië eienskap gebruik.

$$y = \log \cos \left( \cos \left( \frac{\pi}{2} \right) \right)$$

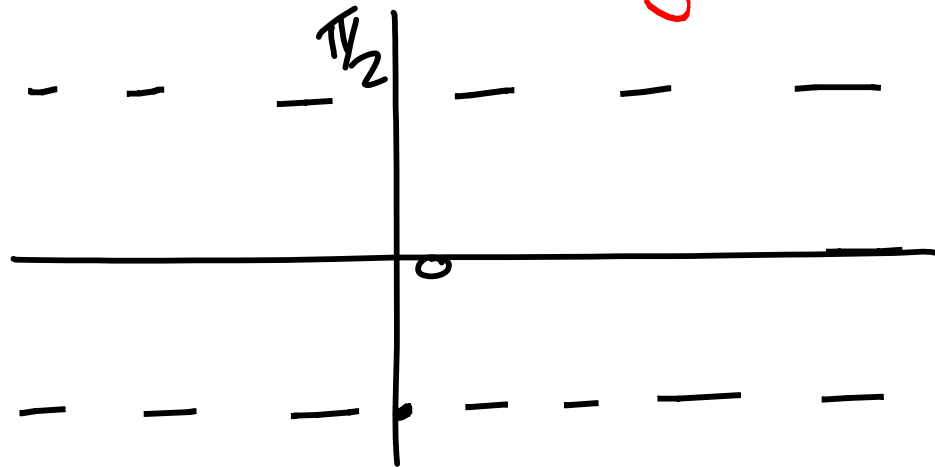
•  $y = \tan x$  • een-een  
duidelijk?

Plan: Laat  $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$   
 $y \in \mathbb{R}$



$y = \text{bg tan } x$

Now is  $x \in \mathbb{R}$  en  $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$



Definitie:  $y = \text{bg tan } x \Leftrightarrow \tan y = x, y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Kansellatie vergelijkingen

$$1) \tan(\text{bg tan } x) = x \text{ as } x$$

$$2) \text{bg tan}(\tan x) = x \text{ as } x$$

Bepaal:

①  $\tan(\lg \tan 10) = x$

② Bepaal zonder inzak rekenaar:

$\tan x = -1$

$$\textcircled{3} \quad y = \log \tan(-1)$$

$$\textcircled{4} \quad \cos\left(\log \tan\left(-\frac{1}{e}\right)\right)$$

$$\textcircled{5} \quad \tan(\arcsin 0.5) = x$$

$$\textcircled{6} \quad \text{Bew } \sin \quad \text{t} \quad \cos(\sin^{-1} x) = \sqrt{1-x^2}$$

⑦ Vereenvoudig  
 $\tan(\sin^{-1}x)$

⑧ Bepaal  $Dg$  en  $Wg$  as  $\bullet g(x) = \sin^{-1}(3x+1)$

Definitievers. van  $y = \sin^{-1}(x)$ :

Now  $g(x) = \sin^{-1}(3x+1)$

$$D_g =$$

$$* W_g = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

(uit definitie van  
 $y = \sin^{-1}x$ )

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MaaK die beste gebruik van elke  
geleentheid = Doen die rechte ding  
op die rechte tyd.

$$* \cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right)$$

Los  $x$  qd:

$$\ln(2x^2 + x) \leq 0$$